

Solution:

$$\text{We know } P(Y=k|X=x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

Note that $\sum_{l=1}^K \pi_l f_l(x)$ is the same for all classes 'k'.

$$\text{We assume } f_k(x) = \frac{1}{\sqrt{2\pi} \sigma_k} e^{-\left[\frac{1}{2\sigma_k^2} (x-\mu_k)^2\right]} \quad (f_k(x) \text{ is Gaussian})$$

$$X_k \sim \mathcal{N}(\mu_k, \sigma_k^2)$$

$$\text{and } \sigma_i^2 = \sigma_j^2 = \sigma^2 \quad \forall i, j \in \{1, 2, \dots, K\}$$

$$\therefore f_k(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\left[\frac{1}{2\sigma^2} (x-\mu_k)^2\right]} \quad \text{and hence,}$$

$$P(Y=k|X=x) = \frac{\pi_k \frac{1}{\sqrt{2\pi} \sigma} e^{-\left[\frac{1}{2\sigma^2} (x-\mu_k)^2\right]}}{\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi} \sigma} e^{-\left[\frac{1}{2\sigma^2} (x-\mu_l)^2\right]}}$$

Taking the logarithm of $P(Y=k|X=x)$ yields

$$\begin{aligned} \ln(P(Y=k|X=x)) &= \ln(\pi_k) - \frac{1}{2\sigma^2}(x-\mu_k)^2 - \ln(\sqrt{2\pi}\sigma) \\ &\quad - \ln\left(\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} e^{-\left[\frac{1}{2\sigma^2}(x-\mu_l)^2\right]}\right) \\ &= \ln(\pi_k) - \frac{1}{2\sigma^2}(x^2 + \mu_k^2 - 2\mu_k x) - \ln(\sqrt{2\pi}\sigma) \\ &\quad - \ln\left(\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} e^{-\left[\frac{1}{2\sigma^2}(x-\mu_l)^2\right]}\right) \end{aligned}$$

Now for every 'k', $\frac{x^2}{2\sigma^2}$ has no dependence on k and

$$\ln\left(\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} e^{-\left[\frac{1}{2\sigma^2}(x-\mu_l)^2\right]}\right) \text{ is the same for all classes 'k'}$$

$\therefore$ to maximize $P(Y=k|X=x)$, we need to maximize the terms directly dependent on 'k', which implies we need to maximize

$$\ln(\pi_k) + \frac{\mu_k x}{2\sigma^2} - \frac{\mu_k^2}{2\sigma^2}$$

Note: As $\ln(\cdot)$ is an increasing function, if we maximize $P(Y=k|X=x)$, we maximize $\ln(P(Y=k|X=x))$.
 $\arg \max_k f_k(x) = \arg \max_k \ln(f_k(x))$

which is the same as maximizing 'k' for the discriminant

$$\text{function } \delta_k(x) = \ln(\pi_k) + \frac{\mu_k x}{2\sigma^2} - \frac{\mu_k^2}{2\sigma^2} \quad (\text{Proved})$$

If $\sigma_i^2 \neq \sigma_j^2 \quad \forall i, j \in \{1, 2, \dots, K\}$, this means that σ_k will depend on different classes. Therefore we now need to maximize $f_k^{\text{QDA}}(x) = \ln\left(\frac{\pi_k}{\sqrt{2\pi}\sigma_k}\right) - \frac{1}{2\sigma_k^2}(x - \mu_k)^2$ which is

a quadratic function in x . In a more formal sense

$$\arg \max_k P(Y=k|X=x) = \arg \max_k \ln(P(Y=k|X=x))$$

$$= \arg \max_k \ln\left(\frac{\pi_k}{\sqrt{2\pi}\sigma_k}\right) - \frac{1}{2\sigma_k^2}(x - \mu_k)^2 = \arg \max_k f_k^{\text{QDA}}(x). \quad \underline{\underline{\text{Proved}}}$$